HHTE — a process-based framework for wetland assessment

Hydrology → Habitat → Trophic organization → Ecosystem functioning

ADAPTIVE MANAGEMENT FEEDBACK

AVOID — **MINIMIZE** — **RESTORE** — **MONITOR**

acts on pressures

PRESSURES

anthropogenic + climatic drivers

- Dams, flow regulation
- Water diversion, pumping
- Drainage, reclamation
- Land-use change
- Climate: rainfall extremes, sea-level rise

HYDROLOGY **HABITAT** **TROPHIC LINKS** **ECOSYSTEM FUNCTIONING**

HVI hydroperiod variability

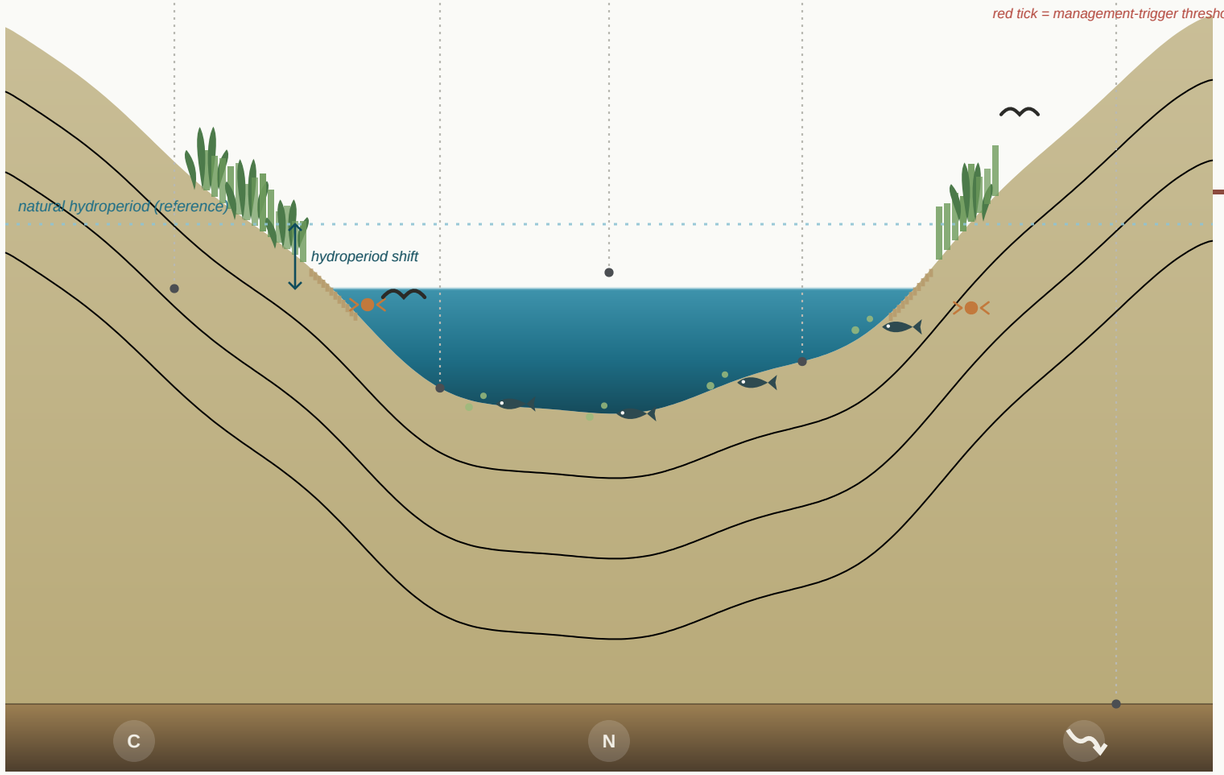
HTR habitat turnover

CI functional connectivity

TRP trophic redundancy

EFS eco-hydrological feedback strength

red tick = management-trigger threshold



ECOSYSTEM FUNCTIONING — carbon storage · nutrient retention · flood regulation

ECOLOGICAL CONSEQUENCES

- Habitat simplification, loss of ecotones
- Reduced trophic redundancy
- Resource–consumer timing mismatch
- Waterbird decline across flyways
- Reduced C storage, nutrient retention, flood-water buffering

Process-based indicators (HVI · HTR · CI · TRP · EFS) are compared with management triggers to activate an avoid – minimize – restore – monitor response.